What is complete, and what is still pending

This is an interim validation snapshot. It freezes plan revision 140 at 2026-11-08 00:43:16 UTC. Later live progress is outside this report. No new main heldout-100 evaluation has run in this continuation campaign.

Closed and reconciled work	Actual amount
Native selection-validation + fresh-confirmation episode executions	23,540
External native action rows / PointEnv transitions	12,793,537
Closed scientific worker time	24,082.604986 seconds (6.690 hours), parent clocks once
New main heldout episodes	0; target remains 100 per selected condition and model

The 23,540 executions span many configurations, tasks and probes. Reset contexts are paired or reused within declared roles; this is not 23,540 independent starting states or trained models.

The clearest completed results

Study	Fresh confirmation	Scope
Task 5 expanded original QGF	40/40, 36/40, 40/40	Seeds 11, 23, 37; own frozen M/kappa choices
Task 2 original QGF	40/40 for all three models	Joint M/kappa selection, then fresh40
Task 2 original FK	40/40 for all three models	Own declared M/N/kappa/noise selection
Task 2 eight DF conditions	Seven seed-11 choices complete	Eighth seed-11 VAL running; later-model VAL/CONF pending

Task 2 seed 11 stochastic/raw and stochastic/averaged-Q lambda 0 both confirm 40/40 at kappa 5, eta .1 and I128. There is no success gain at this ceiling. Regular/averaged lambda 0 confirms 38/40; regular/raw confirms 35/40 with different selected settings.

Interpretation: these are selected-policy validation results. They do not establish an algorithm winner, broad critic repair, or a deterministic advantage at matched settings. Failed IPG engineering gates still block the wider test barrier. [E1-E6]

Tasks and the frozen learned score

The robot is a simulated point controller navigating a MuJoCo maze. Each executed action produces a real next state. This is sequential navigation, rather than a cube-grasp or one-state synthetic task.

Task	Nominal start -> goal	Role in this snapshot
Task 1	(0, 0) -> (20, 20)	Separate FULL/HOLE and right-start coverage probes
Task 2	(0, 20) -> (20, 0)	Current DF / QGF / original particle comparison
Task 5	(20, 4) -> (0, 0)	Own expanded M/kappa QGF validation and confirmation

These are nominal maze coordinates; native start noise is retained. The native cap is 1,000 environment steps. Position control resets velocity each step, and simulator/wrapper/RNG state is checkpointed for resume. The right-start Task 1 probe is a separately labeled custom reset distribution. [E7]

Behavior flow, Q and V stay fixed

The actor is the offline behavior-cloning flow checkpoint at step 1,000. Goal-specific IQL critic/target-critic/value checkpoints are from step 50,000. Each training seed has its own frozen model. The same actor and critic are reused across that model's sampler comparisons; no online critic learning is added.

Actual score in code: agent._aggregate_q(agent.target_critic(s,a)). All nine actor / Task 2 / Task 5 config files for seeds 11, 23 and 37 specify **q_aggregation = min** with two heads. Raw Q is therefore the minimum learned IQL target-Q prediction, without action-neighborhood averaging. It is not an oracle return. [E7]

```
IQL target: y = reward + 0.99^H * mask * V(next_state); H = 1
```

V is fit by expectile regression against aggregated target Q (expectile .8). The target critic update coefficient is .005. An IQL/Bellman target or self-consistency check alone is not an independent estimate of true environment Q. [E7]

The evaluation roles

Role	Purpose
TRAIN16	Fixed paired data states for the declared DF/FK hierarchy
VAL20	Compare the complete declared native shortlist on paired contexts
Fresh40 confirmation	Evaluate literal top three; freeze a choice using fresh40 only
Main heldout100	After all required methods/models freeze; still unrun here

Methods and computation: M, I, N and K

Symbol	Meaning
M	Flow integration slices per action. QGF searches 8, 16, 32 and 64. The actor was trained with eight slices.
I	DF source / latent updates before one physical action; current DF endpoints remain full F8.
N	Particle count for original IPG/FK. One environment action is selected from the final population.
K	Critic-neighborhood action samples for Q averaging: total 4 or 16.
100	Future heldout episodes per selected condition and trained model; it is not a particle count.

Identity-gradient, full-endpoint D-Flow

```
a_i = full_F8(s,z_i); g_i = action_gradient Q_or_Qbar(s,a_i)
```

```
z_next = z + eta*(kappa*g_i - lambda*z_i)
```

```
Stochastic condition adds sqrt(2*eta*T)*xi; T = 1
```

The endpoint uses all eight flow slices. The backward endpoint Jacobian is replaced by the identity: this is a surrogate gradient, not the full Jacobian through F8. Regular DF still has a Gaussian initial source z0. Stochastic DF adds fresh source-update noise. Lambda 0 and 1 are separate prior-pull settings; lambda 1 is a Gaussian quadratic prior term, **not an L1 penalty**.

Original QGF keeps its per-slice proxy

At each t = i/M, QGF queries its clean-action proxy clip(a_t + (1-t)*v_BC, -1, 1), takes the action-Q gradient, and updates the vector field. Its original one-Euler proxy is retained. This is separate from DF's full-F8 endpoint. Increasing M changes inference integration, without retraining the F8 actor.

Averaged Q uses the same frozen critic

```
Qbar_K(s,a) = (1/K) * sum_k min_j Qtarget_j(s, clip(a + sigma*eps_k,-1,1))
```

The antithetic standard-normal epsilon bank is fixed through all I updates of one physical decision and refreshed for the next decision. Offsets are generated around the candidate action, not retrieved dataset actions or new observed data. Sigma .03 is standard deviation .03 per coordinate before clipping, not a hard radius. Finite K is a bank-conditioned score, not an exact Gaussian expectation. [E7]

Task 2: all eight DF conditions

Each row below is one condition from four DF families crossed with fixed lambda 0/1. Seven seed-11 choices are frozen from fresh40. No seed-23 or seed-37 native selection outcome is filled in from another model. [E2]

Condition	lambda	Seed 11 selected	sigma / K	CONF	Seed 23	Seed 37
Regular / raw Q	0	k=0.1; eta=0.1; I256	-	35/40	TRAIN done; VAL/CONF pending	TRAIN pending
Regular / raw Q	1	k=40; eta=0.03; I256	-	28/40	TRAIN done; VAL/CONF pending	TRAIN pending
Stochastic / raw Q	0	k=5; eta=0.1; I128	-	40/40	TRAIN done; VAL/CONF pending	TRAIN pending
Stochastic / raw Q	1	k=5; eta=0.1; I128	-	37/40	TRAIN done; VAL/CONF pending	TRAIN pending
Regular / averaged Q	0	k=1; eta=0.1; I256	s=0.003; K16	38/40	TRAIN done; VAL/CONF pending	TRAIN pending
Regular / averaged Q	1	k=64; eta=0.03; I256	s=0.5; K4	25/40	TRAIN done; VAL/CONF pending	TRAIN pending
Stochastic / averaged Q	0	k=5; eta=0.1; I128	s=0.003; K16	40/40	TRAIN done; VAL/CONF pending	TRAIN pending
Stochastic / averaged Q	1	pending	-	pending	TRAIN done; VAL/CONF pending	TRAIN pending

For seed 11 stochastic/averaged lambda 1, VAL22 is running at this snapshot and fresh40 is pending. A pending cell has no completed success denominator; it is not zero. Each selected configuration and pending status for all models is exported in the companion settings CSV.

The declared hierarchy has not been reduced

DF kappa: 0, .02, .04, .08, .1, .2, .4, .8, 1, 2, 3, 5, 8, 12, 16, 20, 32, 40, 64, 80.
Eta: .001, .003, .005, .01, .03, .1. I: 1, 2, 4, 8, 16, 32, 64, 128, 256.
Averaged-Q sigma: .003, .01, .03, .06, .1, .2, .3, .5. K: 4 or 16.

The predeclared hierarchy is at most 9,240 TRAIN settings/model, rather than all 73,440 Cartesian combinations. Seeds 11 and 23 completed their eight TRAIN grids; seed 37 is pending. Raw families receive seven native shortlist requests; averaged families receive 22, including the shared canonical BC control where exact cache identity qualifies. [E2]

Only the literal overall top three receive fresh40. No hidden fourth positive candidate replaces a zero or a failed outcome. Current healthy DF engineering is scoped to I at most 256; the older I512 failure is retained.

QGF and original FK: fresh40 settings

Task 5: retain every one of the nine confirmation outcomes. All 324 selection batches (27 kappa values x four M values x three models) were new executions. Their literal top three/model then received 360 fresh confirmation episodes. [E3]

Model seed	kappa	M	Fresh40	Selection status
11	256	64	40/40	other literal candidate
11	96	16	40/40	frozen choice
11	32	64	39/40	other literal candidate
23	5	64	32/40	other literal candidate
23	2	64	36/40	frozen choice
23	1	64	35/40	other literal candidate
37	96	32	40/40	other literal candidate
37	80	32	40/40	frozen choice
37	128	32	40/40	other literal candidate

Task 5 frozen choices are kappa96/M16, kappa2/M64 and kappa80/M32 for seeds 11, 23 and 37. All three are also the positive-guidance choices; they are views of the same episodes, not extra trials. Ties use the original return/distance/cost ranking. These are confirmation results, not heldout100.

Task 2: three QGF choices and three FK choices

Seed	Method	kappa	Own settings	Fresh40
11	Original QGF	16	M64	40/40
11	Original FK	80	M64; N32; noise 0.02	40/40
23	Original QGF	320	M8	40/40
23	Original FK	80	M32; N32; noise 0.02	40/40
37	Original QGF	16	M16	40/40
37	Original FK	80	M16; N32; noise 0.02	40/40

QGF searched 27 kappa values through 320 and M8/16/32/64. Original FK used 720 TRAIN settings/model over its original strength grid through 80, M8/16/32/64, N8/16/32 and declared noise .0/.005/.02. FK3 and QGF3 freezes alone do not open the all-method test barrier. [E4]

What can be said about averaging and noise

Averaging changes the objective that guides the same frozen policy/critic pair. It does not add experience, retrain Q, or automatically make an unsupported action accurate. Near action bounds, clipping the perturbations can alter the local gradient; larger sigma is not automatically better.

Own selected policies: fresh40, Task 2 seed 11

Mode / lambda	Raw Q selected	Averaged Q selected	Reading
Regular / 0	35/40; k .1, eta .1, I256	38/40; k 1, eta .1, I256; s .003/K16	Different selected kappa
Regular / 1	28/40; k 40, eta .03, I256	25/40; k 64, eta .03, I256; s .5/K4	Different selected kappa and score
Stochastic / 0	40/40; k 5, eta .1, I128	40/40; k 5, eta .1, I128; s .003/K16	Same core settings; success ceiling
Stochastic / 1	37/40; k 5, eta .1, I128	pending	No completed averaged comparison

These comparisons summarize independently selected policies. They do not isolate a general averaging effect. The stochastic lambda 0 pair shares core settings and reaches the same 40/40 ceiling; that does not establish broad equivalence or negative averaging effects.

Existing matched raw-DF VAL20 overlaps

lambda	kappa	eta	I	Regular	Stochastic
0	1	0.1	256	13/20	19/20
1	1	0.1	128	0/20	5/20
1	5	0.1	128	18/20	19/20

These three rows are all literal raw-DF mode-to-mode overlaps in the four completed seed-11 VAL20 lists. They share paired starts and exact kappa/eta/I; shared BC control references are excluded from this mode comparison. This is a posthoc, one-model validation description, not a full matched heldout curve. [E5]

The planned matched-kappa study and own-best tuned curves answer different questions. Equal kappa across QGF and DF does not ensure equal accumulated guidance because M, I, eta, the endpoint/proxy and objective differ. Current results do not establish deterministic superiority at matched settings.

Verification failures and the IPG finding

IPG is still unqualified for native selection in this expanded study. Saved diagnostics narrow a numerical question; they do not waive the original gates or introduce another tested policy. [E6]

Retained evidence	What actually happened
Original particle engineering	33 controlled IPG solve-quality failures; 15 learned IPG cases remained missing. FK was completed and qualified separately.
Mixed-precision verification	Early dtype/lowering failures were preserved. The V4 synthetic gate ran all 144 cases: 134 PASS and 10 FAIL; no learned Stage C or native IPG selection followed.
Captured-operand diagnostics	Exported A/RHS/phi and saved comparisons were studied separately. Old public-output parity is credited to the prior capture audit, not claimed as new replay.
Isolated saved-system solve	40 B2 calls / 80 systems; three dtype proofs; no NA. Maximum difference from saved HOST solution: 1.7053e-12. Difference from captured in-graph phi reached .0003135583.

The isolated solver ran in a different compilation context. Close HOST agreement does not establish the hidden original solver-consumer bits, a unique compiler cause, or a repaired IPG method. The 80 systems and 160 saved comparison records are offline diagnostics, not new navigation episodes.

Other boundaries remain explicit

Boundary	Current scope
Task 5 cross-shape check	The original 72-cell gate retained one B2/B20 action mismatch above 2e-6. A separate 324-cell canonical-B20 gate passed; this is a narrower contract, not a universal batch/state/RNG guarantee.
TRAIN critic-shape probe	The initial config guard failed before score queries. The repaired guard/proof-persistence stage completed 540 TRAIN score/gradient records; these are not Q truth or heldout results.
Unseen-action problem design	Coverage exposure and critic-shape probes do not force a QGF failure. Harder spurious-peak, broad-error and precision-control studies remain distinct proposals.

The custom right-start coverage probe found all six QGF validation cells could reach 20/20 at their best strengths. Exposure to the withheld action region is not itself evidence of incorrect Q ranking or a failure mode. No such result is relabeled as proof that averaging repairs unseen Q.

Pending work and reproducible evidence

What remains before the new heldout100 study

Branch	Status fixed at plan 140
Task 2 D-Flow	Finish seed-11 stochastic/averaged lambda1 VAL and fresh40; then the declared remaining model phases. Seed23 TRAIN is complete; seed37 TRAIN is pending.
Original IPG	Engineering remains unqualified. No native selection or test can be filled in from a diagnostic.
All-method freeze	Eight DF conditions + QGF + IPG + FK = 11 guided choices/model, 33 across three models. This full barrier is not complete.
Matched-setting curves	Separate proposed matched-kappa/eta/I evaluation remains pending. No heldout results are used to tune current candidates.
Task 5 saved traces / graph artifacts	Source preparation only in this snapshot. No later plotting or compiler-artifact result is assumed here.

Read the saved evidence with the report

Ref	Frozen sources
E1	Plan140 exact copy and reconciled counts; DATA.json. Plan SHA starts 38c9702f.
E2	Seven DF seed11 selection files, fresh40 metrics/raw CSV + metadata and independent phase audits.
E3	Task5 all324 audit fd382fe3; fresh40 audit 4f10cd7e; marker 3a1786af; all nine outcomes CSV.
E4	Task2 QGF fresh40 audit 2a6b738f; FK all-three selection files and independent confirmation audits.
E5	Exact matched raw-DF overlap JSON and independent review; all three overlaps retained.
E6	IPG V4 failed-gate audit and isolated-system saved audit 127e1477; no qualification.
E7	Frozen qgf.py / agents/common.py / sampler and full-endpoint code, nine min-Q configs, native engine and maze/point sources.

Companions: ALL_MODELS_SELECTION_STATUS_AND_SETTINGS.csv has exact selected configs and pending rows for all three models, including full frozen config JSON and source hashes. TASK5_ALL9_FRESH40_OUTCOMES.csv retains all nine candidates. DATA.json stores the reported scalars; EVIDENCE_MANIFEST.json identifies copied evidence and opaque trace references.

This is an immutable interim snapshot, not the prior morning report, a full historical archive or a new main-test completion claim. Older heldout results remain separate history. No arrays, models, samplers, environments or RNGs were queried to author this report.